

John Doe

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EDUCATION

Program	Institution	Grade/Percentage	Completion
B.Tech, Computer Science and Engineering	Indian Institute of Technology Madras		2021
12th Std. PCMC (CBSE)	XYZ Public School		2017
10th Std. (CBSE)	XYZ Public School		2015

WORK EXPERIENCE

Smart Order Router Analytics

Summer Internship

May-July 2020

XYZ Company

- Improved the performance of an order router by analysing market data and factoring in cross market differences.
- Used Q/KDB+ for interacting with databases and R for plotting. Work was done on a remote host.

PROJECTS

Frequent Item Set Mining on Live Twitter Feed

CS6847, Cloud Computing

Jan-May 2020

IIT Madras

- Mined frequent item sets using Spark simultaneously across two cloud instances (Amazon-WS EC2).
- Kafka was used to keep the back end distributed, while the Zookeeper configuration service managed the cluster.

Image Captioning and Machine Translation

CS6910, Deep Learning

Jan-May 2020

IIT Madras

- Captioned images via a CNN (netVLAD) encoder and LSTM decoder architecture, written in Tensorflow and Keras.
- Translated English sentences to Hindi using a supervised transformer model with multi-head attention, written in Pytorch.

Map-Reduce on Hadoop Cluster

CS6847, Cloud Computing

Jan-May 2020

IIT Madras

- Calculated the most common travel routes (of a car rental service) using a three node Hadoop cluster.
- The problem was phrased as a map-reduce task, and performance was compared as the number of reducers were varied.

Covert Channel Communication through TLB

CS3700, Operating Systems

Jul-Nov 2019

IIT Madras

- Exploited the translation look-aside buffer (TLB) of a PC to allow two processes to communicate without OS permission.
- Memory access times were measured to determine TLB hits/misses, controlled by one process flushing the TLB.
- Second process reads 0/1 based on the state of the TLB, allowing transmission of coarse grained information.

Spoken and Written Character Recognition

CS5691, Pattern Recognition and ML

Jul-Nov 2019

IIT Madras

- Performed recognition on written Telugu characters, via a Hidden Markov Model (HMM) and Dynamic Time Warping (DTW)
- Character recognition performed on spectral frequency coefficients of recorded spoken Telugu consonants, using HMMs.

Fraudulent Credit Card Detection

CH5019, Mathematics for Data Science

Jan-May 2019

IIT Madras

- Compared the performance of non-linear kernels in Support Vector Machines in detecting defaulting customers from a bank dataset using python (pandas, scikit-learn)

Sports Ranking Predictor

MS4110, Data Analytics

July-Nov 2018

IITM

- Built a time series (ARIMA) model to predict sports teams rankings using python (pandas, scikit-learn)

COURSEWORK

Computer Science

Cloud Computing	Approximation Algorithms	Advanced Graph Algorithms*	Blockchain & Distributed Technologies*
Cryptography*	Randomized Algorithms*	Design and Analysis of Algorithms	Languages Machines & Computation
Database Systems	Operating systems	Paradigms of Programming	Computer Organization & Architecture
Compilers	Computer Networks	Object Oriented Programming	

Machine Learning/Data Science

Deep Learning Pattern Recognition and ML Data Analytics Mathematics for Data Science

Mathematics

Discrete Mathematics Probability and Statistics Graph Theory

** Ongoing courses*

SKILLS

Languages : C/C++ • Python • Java • Q/KDB+ • Scala • SQL • Prolog • OCaml • Flex + Bison • R • LaTeX • Bash

Frameworks/Libraries : Hadoop • Keras and Tensorflow • Pytorch • Numpy and SciKit Learn • Kafka • Spark • Zookeeper • Git, Github

ACADEMIC ACHIEVEMENTS

2017 All India Rank — in JEE Advanced
2016 Kishore Vaigyanik Protsahan Yojana (KVPY) Scholar
2015 National Talent Search Examination (NTSE) Scholar

CO&EXTRA-CURRICULARS

2018-19 Represented IIT Madras in basketball tournaments
2017 President of school Mathematics Association. Organized inter-school Mathematics Fest
2016 Secretary of school Computer Science Association